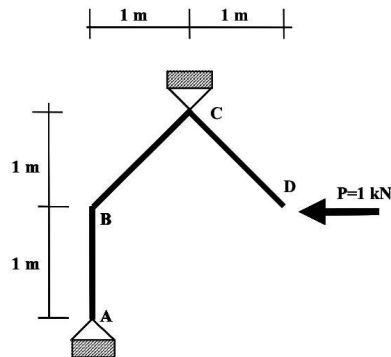
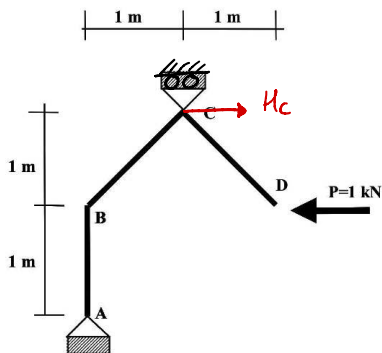


PROBLEMA 1.2: La viga continua de la figura está sometida a una carga puntual $P=1\text{ kN}$ en el nudo D con la dirección y sentido indicados en la figura. Sabiendo que el producto EI de todas las barras toma el valor $EI=100\text{ kN}\cdot\text{m}^2$, se pide calcular:

- El grado de hiperestatismo externo de la estructura indicando el método a seguir para resolver el problema.
- Las reacciones en los apoyos A y C.
- Los diagramas de momentos flectores de la estructura.
- Los movimientos de los puntos A y C.

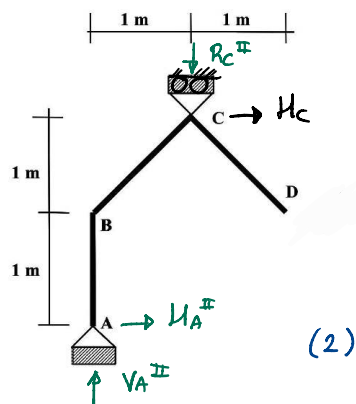
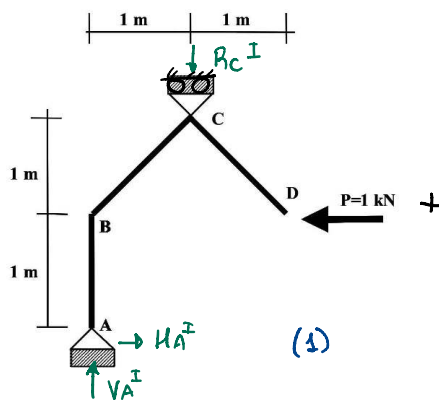


$$G_U = G_{HE} = \underset{\substack{\uparrow \\ \text{reacciones} \\ \text{externas}}}{CE} - GDLE = 4 - 3 = 1$$



$$V_C = 0$$

Superposición :



(1)

$$\sum F_v = 0 \Rightarrow V_A^I = R_C^I = 1 \text{ kN}$$

$$\sum F_h = 0 \Rightarrow H_A^I = 1 \text{ kN}$$

$$\sum M_A = 0 \Rightarrow 1 \cdot 1 = R_C^I \cdot 1$$

$$R_C^I = 1 \text{ kN}$$

(2)

$$\sum F_v = 0 \Rightarrow V_A^{II} = R_C^{II} = -2 H_c$$

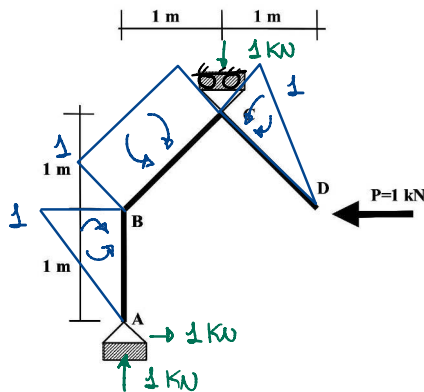
$$\sum F_h = 0 \Rightarrow H_A^{II} = -H_c$$

$$\sum M_A = 0 \Rightarrow R_C^{II} = -2 H_c$$

Las reacciones totales son la suma de (1) y (2)

Momentos flectores

(1)

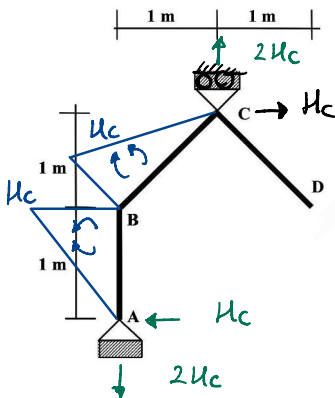


Corte AB: $\uparrow$ $M_F = -1 \cdot x \text{ kNm}$

Corte CD: $\uparrow$ $M_F = -1 \cdot \sin 45^\circ \cdot x$

Corte BC: $\uparrow$ $M_F = -1 + 1 \sin 45^\circ \cdot x - 1 \sin 45^\circ x = 1 \text{ kNm}$

(2)



Para calcular H_c se hace $\vec{U}_c = 0$ y $\vec{V}_c = 0$ desde A

$$\vec{U}_c = \cancel{\vec{U}_A} + \vec{\delta}_A (2) + \frac{1}{EI} \left(\frac{1}{2} \cdot 1 \cdot 1 \cdot \frac{4}{3} + 1 \cdot \sqrt{2} \cdot \frac{1}{2} - \frac{1}{2} H_c \cdot \frac{4}{3} - \frac{1}{2} H_c \cdot \sqrt{2} \cdot \frac{2}{3} \right)$$

$$2 \vec{\delta}_A + \frac{1}{100} (1,37 - 1,1380 H_c) = 0$$

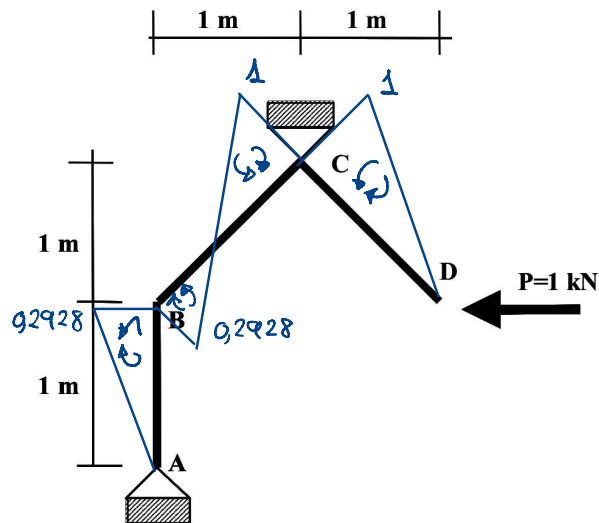
$$\downarrow \cancel{r_C} = \cancel{v_A} + \hat{\theta}_A \cdot 1 + \frac{1}{EI} \left(+ \frac{1}{2} \cdot 1 \cdot 1 \cdot 1 + 1 \cdot \sqrt{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot H_C \cdot 1 \cdot 1 - \frac{1}{2} H_C \cdot \sqrt{2} \cdot \frac{2}{3} \right) = 0$$

$$\hat{\theta}_A + \frac{1}{100} (1,2071 - 0,9714 H_C) = 0$$

$$\hat{\theta}_A = 4,874 \cdot 10^{-4} \text{ rad}$$

$$H_C = 1,2928 \text{ kN}$$

$$(1) + (2) =$$



Movimientos en A y C

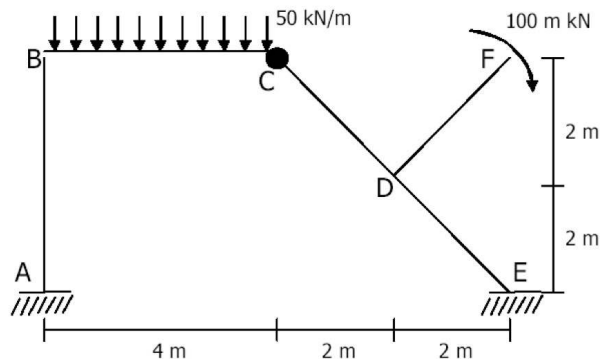
$$\hat{\theta}_A = 4,874 \cdot 10^{-4} \text{ rad}$$

C sólo tiene sino también:

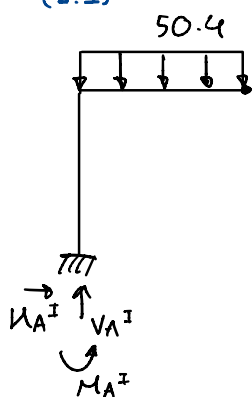
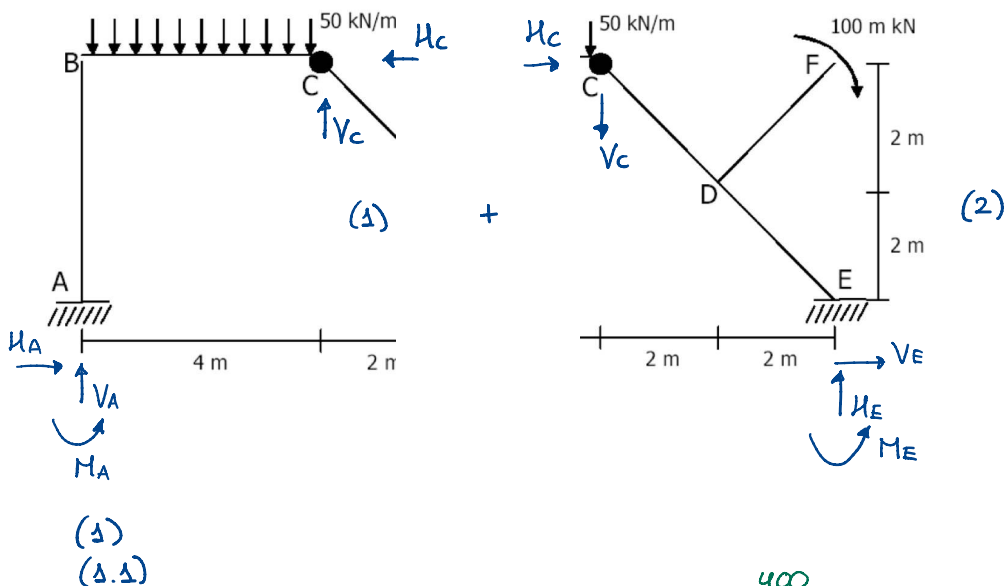
$$\hat{\theta}_C = \hat{\theta}_A + \frac{1}{EI} \left(\frac{1}{2} \cdot 1 + 1 \cdot \sqrt{2} - 1,2928 \cdot \frac{1}{2} - 1,2928 \cdot \frac{\sqrt{2}}{2} \right) = 4,024 \cdot 10^{-3} \text{ rad}$$

PROBLEMA 1.4: En la figura se muestra un croquis acotado de una estructura en la que cabe destacar que el nudo C es una rótula. Las cargas actuantes son: una sobrecarga vertical hacia debajo de 50 kN/m sobre la barra BC y un momento de 100 m·kN en el extremo F de la barra FD. El producto EI es igual a 6000 kN·m² para todas las barras. Se pide:

- Calcular las reacciones en los empotramientos.
- El giro en el nudo F.



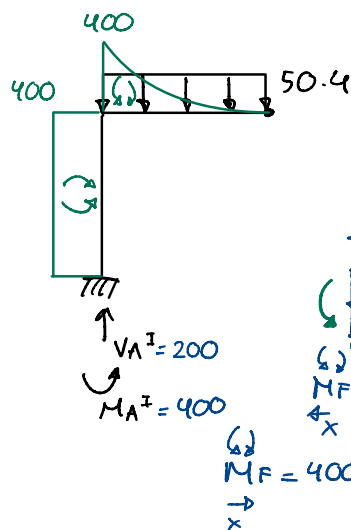
a) $GH = G_{UE} + G_{UI}$
 $G_{UE} = 6 - 3 = 3$ unión AB rótula unión CDFE
 $G_{UI} = CI - G_{LI} = 3(2-1) + 2(2-1) + 3(3-1) - 3(5-1) = -1$
 $GH = 2$ Hiperestática de grado 2



$$V_A^I = 200 \text{ kN}$$

$$H_A^I = 0$$

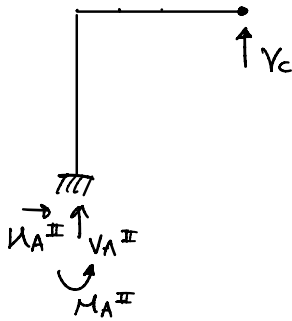
$$M_A^I = 400 \text{ kN}$$



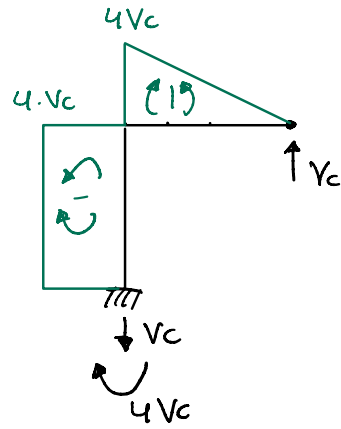
$$M_F = 50 \cdot x \cdot \frac{x}{2} = 25x^2$$

$$M_F = 400 - 25x^2$$

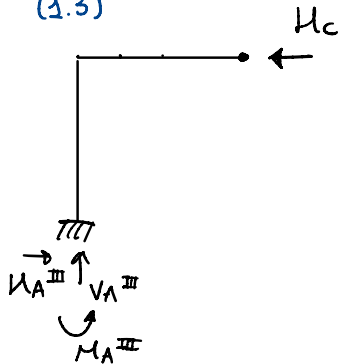
(1.2)



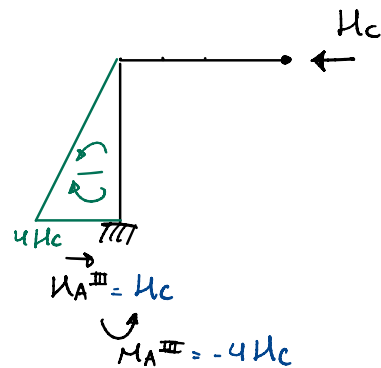
$$\begin{aligned} H_A^{\text{II}} &= 0 \\ V_A^{\text{II}} &= -V_C \\ M_A^{\text{II}} &= -V_C \cdot 4 \end{aligned}$$



(1.3)



$$\begin{aligned} H_A^{\text{III}} &= H_C \\ V_A^{\text{III}} &= 0 \\ M_A^{\text{III}} &= -4 H_C \end{aligned}$$



$$\tilde{U}_{C1} = \cancel{\frac{1}{2} \cdot 0 \cdot 4} + \cancel{0 \cdot 4} + \frac{1}{EI} \left[400 \cdot 4 \cdot 2 - 4 \cdot V_C \cdot 4 \cdot 2 - \frac{1}{2} \cdot 4 H_C \cdot 4 \cdot \frac{8}{3} \right] =$$

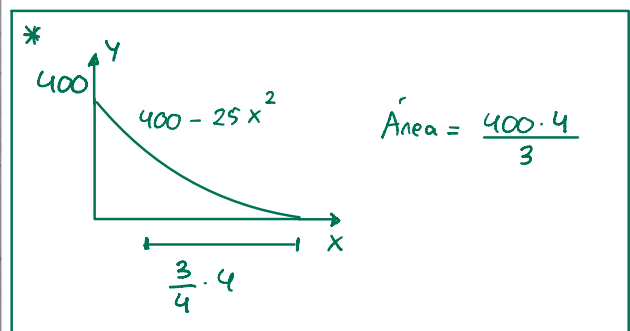
$$= \frac{1}{EI} \left[3200 - 32 V_C - \frac{64}{3} H_C \right]$$

$$\downarrow \tilde{V}_{C1} = \cancel{\frac{1}{2} \cdot 0 \cdot 4} + \cancel{0 \cdot 4} + \frac{1}{EI} \left[400 \cdot 4 \cdot 4 + 400 \cdot \frac{4}{3} \cdot \frac{3}{4} \cdot 4 - 4 V_C \cdot 4 \cdot 4 - \frac{1}{2} \cdot 4 V_C \cdot 4 \cdot \frac{2}{3} \cdot 4 - \right.$$

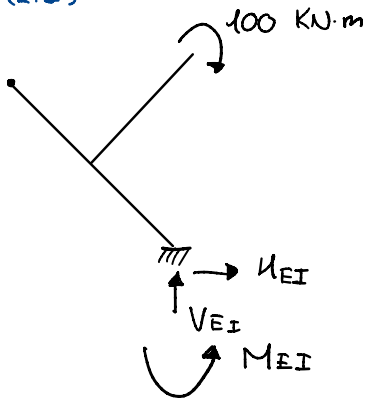
$$\left. - \frac{1}{2} \cdot 4 H_C \cdot 4 \cdot 4 \right] =$$

$$= \frac{1}{EI} \left[8000 - \frac{256}{3} V_C - 32 H_C \right]$$

Shape		$\bar{x}$	$\bar{y}$	Area
Triangular area			$\frac{h}{3}$	$\frac{bh}{2}$
Quarter-circular area		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Semicircular area		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
Quarter-elliptical area		$\frac{4a}{3\pi}$	$\frac{4b}{3\pi}$	$\frac{\pi ab}{4}$
Semielliptical area		0	$\frac{4b}{3\pi}$	$\frac{\pi ab}{2}$
Semiparabolic area		$\frac{3a}{8}$	$\frac{3h}{5}$	$\frac{2ah}{3}$
Parabolic area		0	$\frac{3h}{5}$	$\frac{4ah}{3}$
Parabolic spandrel		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$
General spandrel		$\frac{n+1}{n} a$	$\frac{ah}{4n+2}$	$\frac{ah}{n+1}$
Circular sector		$\frac{2r \sin \alpha}{3\alpha}$	0	αr^2

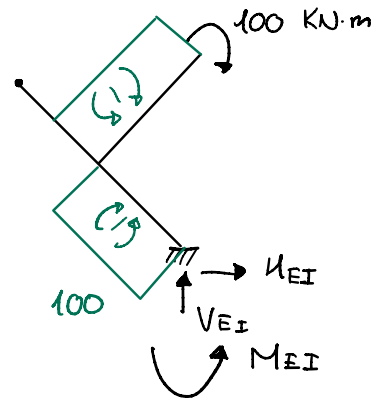


(2)
(2.1)

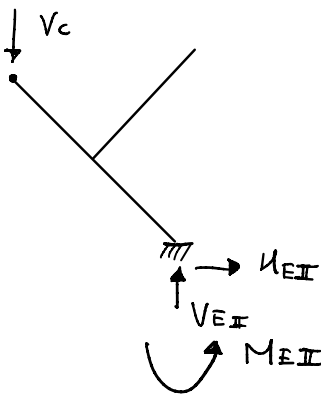


$$V_{EI} = H_{EI} = 0$$

$$M_{EI} = 100 \text{ kNm}$$



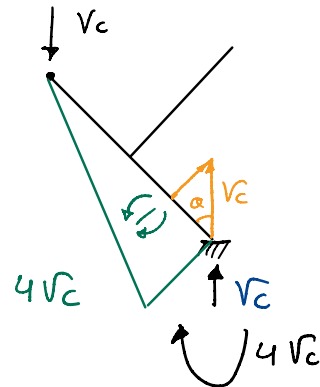
(2.2)



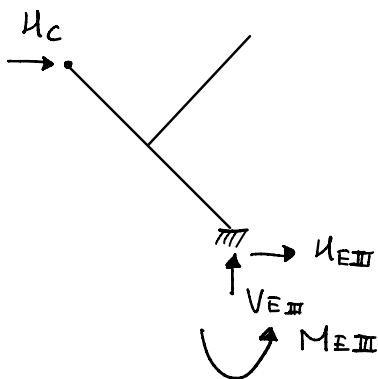
$$V_{EII} = V_c$$

$$H_{EII} = 0$$

$$M_{EII} = -4V_c$$

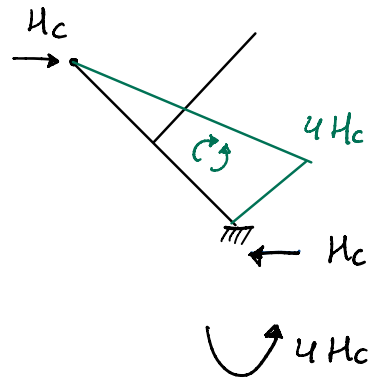


(2.3)



$$H_{EIII} = -H_c$$

$$M_{EIII} = 4H_c$$



$$\vec{u}_{c2} = \vec{u}_E + \vec{\theta}_E \cdot 4 + \frac{1}{EI} \left[100 \cdot 2 \cdot \frac{2}{\sqrt{2}} \cdot 3 - \frac{1}{2} 4V_c \cdot \frac{2}{\sqrt{2}} \cdot 4 \cdot \frac{2}{3} + \frac{1}{2} 4H_c \cdot \frac{2}{\sqrt{2}} \cdot 4 \cdot \frac{2}{3} \cdot 4 \right] = \frac{1}{EI} \left[600\sqrt{2} - \frac{64}{3}\sqrt{2}V_c + \frac{64}{3}\sqrt{2}H_c \right]$$

$$\downarrow v_{c2} = \downarrow v_E + \vec{\theta}_E \cdot 4 + \frac{1}{EI} \left[-100 \cdot 2 \cdot \frac{2}{\sqrt{2}} \cdot 3 + \frac{64}{3}\sqrt{2}V_c - \frac{64}{3}\sqrt{2}H_c \right] = \frac{1}{EI} \left[-600\sqrt{2} + \frac{64}{3}\sqrt{2}V_c - \frac{64}{3}\sqrt{2}H_c \right]$$

$$\vec{u}_{c1} = \vec{u}_{c2} ; \quad \vec{v}_{c1} = \vec{v}_{c2}$$

$$\begin{cases} 3200 - 32 V_c - \frac{64}{3} H_c = 600 \sqrt{2} - \frac{64 \sqrt{2}}{3} V_c + \frac{64 \sqrt{2}}{3} H_c \\ 8000 - \frac{256}{3} V_c - 32 H_c = -600 \sqrt{2} + \frac{64 \sqrt{2}}{3} V_c - \frac{64 \sqrt{2}}{3} H_c \end{cases}$$

$$\begin{cases} -51,5032 H_c - 1,83011 V_c = -2351,4718 \\ -1,8301 H_c - 115,503 V_c = -8848,5281 \end{cases} \Rightarrow$$

$$\begin{aligned} H_c &= 42,96 \text{ kN} \\ V_c &= 75,93 \text{ kN} \end{aligned}$$

Reacciones :

$$\vec{H}_A = H_A^I + H_A^{II} + H_A^{III} = H_c = 42,96 \text{ kN}$$

$$\vec{V}_A = V_A^I + V_A^{II} + V_A^{III} = 200 - V_c = 124,07 \text{ kN}$$

$$\vec{M}_A = M_A^I + M_A^{II} + M_A^{III} = 400 - 4 \cdot V_c - 4 \cdot H_c = -75,56 \text{ kNm}$$

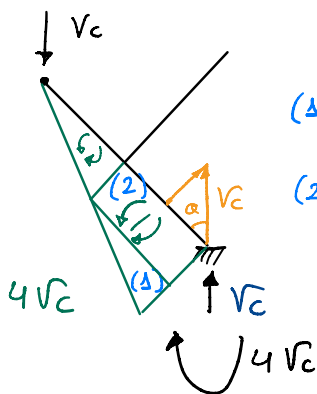
$$\vec{V}_E = V_c = 75,93 \text{ kN}$$

$$\vec{H}_E = -H_c = -42,96 \text{ kN}$$

$$\vec{M}_E = 100 - 4 \cdot V_c + 4 \cdot H_c = -31,88 \text{ kNm}$$

b) Giro en F

$$\begin{aligned} \hat{\theta}_F &= \hat{\theta}_E + \frac{1}{EI} \left[100 \cdot 2 \cdot \sqrt{2} + 100 \cdot 2 \cdot \sqrt{2} - \underbrace{2 V_c \sqrt{2}}_{(1)} - \underbrace{4 V_c \sqrt{2}}_{(2)} + 2 H_c \sqrt{2} + 4 H_c \sqrt{2} \right] = \\ &= 90476 \text{ rad} \end{aligned}$$

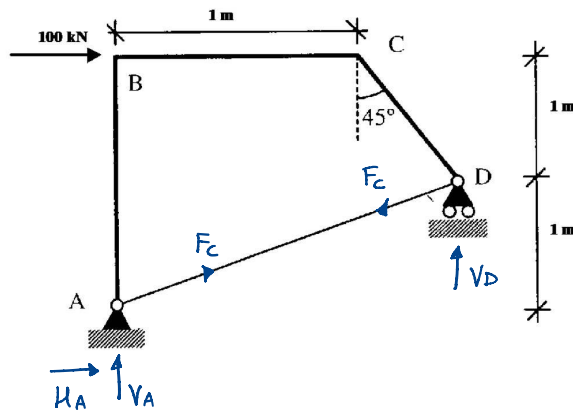


$$(1) \quad \frac{1}{2} \cdot 2 V_c \cdot 2 \cdot \sqrt{2} = 2 V_c \sqrt{2}$$

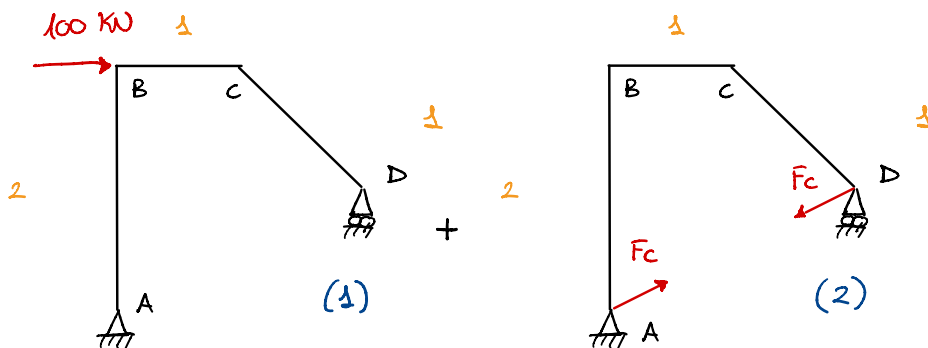
$$(2) \quad 2 V_c \cdot 2 \sqrt{2} = 4 V_c \sqrt{2}$$

PROBLEMA 1.6: Las piezas prismáticas del pórtico de la Figura tienen una rigidez $EI=30000$ $\text{kN}\cdot\text{m}^2$. Los puntos A y D están unidos por un cable flexible con $EA=4000$ kN. Determinar:

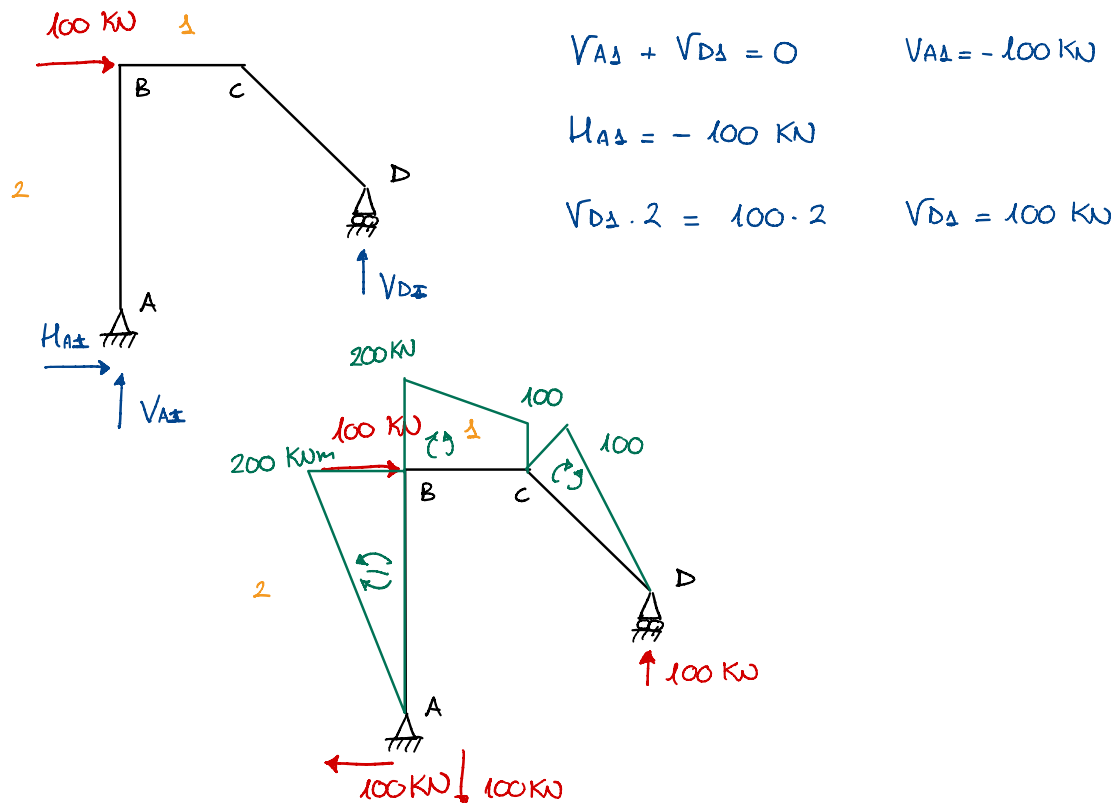
- La fuerza en cable
- Las leyes de esfuerzos



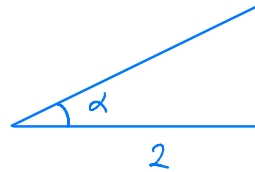
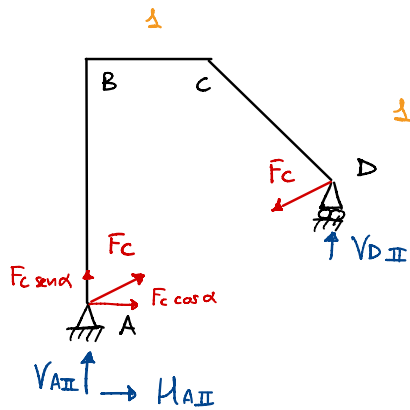
$GHE = 3 - 3 = 0$
 $GHI = 3(2-1) \cdot 2 - 3(3-1) = 0$
 $GH = 0$ isostática
 Como no se conoce F_c tenemos 4 incógnitas y 3 ecuaciones
 $\Downarrow$
 Superación



(1)



(2)



$$\alpha = \arctan \frac{1}{2} = 26,565^\circ$$

$$\sum F_v = 0$$

$$\cancel{F_c \sin \alpha} + V_{DII} + V_{AII} = \cancel{F_c \sin \alpha}$$

$$\sum F_H = 0$$

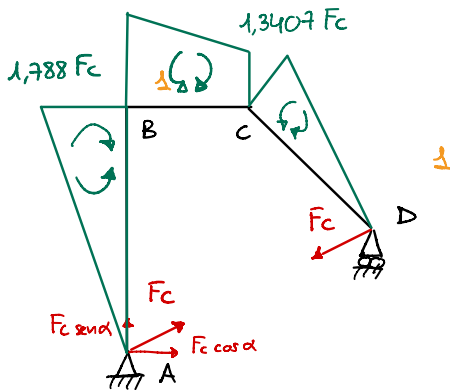
$$H_{AII} + F_c \cos \alpha - F_c \cos \alpha = 0$$

$$\sum \bar{M}_A = 0$$

$$2 V_{DII} + F_c \cdot \cos \alpha = F_c \sin \alpha \cdot 2$$

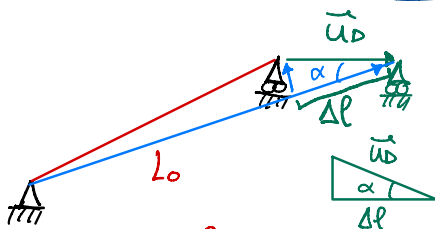
$$V_{DII} = F_c \left(\sin \alpha - \frac{\cos \alpha}{2} \right) = 0$$

$$V_{AII} = 0$$



$$\begin{aligned} V_D = \cancel{V_A} + \bar{\theta}_A \cdot 2 + \frac{1}{EI} \left[-\frac{1}{2} \cdot 200 \cdot 2 \cdot 2 - 100 \cdot 1 \cdot \frac{3}{2} - \frac{1}{2} \cdot 100 \cdot 1 \cdot \left(1 + \frac{2}{3}\right) - \right. \\ \left. - \frac{1}{2} \cdot 100 \cdot 1 \cdot \sqrt{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot 1,788 F_c \cdot 2 \cdot 2 + 1,3407 F_c \cdot 1 \cdot \frac{3}{2} + \frac{1}{2} \cdot 0,4473 F_c \cdot 1 \cdot \left(1 + \frac{2}{3}\right) + \right. \\ \left. + \frac{1}{2} \cdot 1,3407 F_c \cdot \sqrt{2} \cdot \frac{2}{3} \right] \\ 2 \bar{\theta}_A + 2,197 \cdot 10^{-4} F_c - 0,0227 = 0 \quad (1) \end{aligned}$$

$$F = EA \frac{\Delta l}{L_0} \Rightarrow F_c = EA \cdot \bar{u}_D \frac{\cos^2(26,565)}{2} \Rightarrow \bar{u}_D = 2 \frac{F_c}{EA} \frac{1}{\cos^2(26,565)} \quad (2)$$



$$\Rightarrow L_0 = \frac{2}{\cos(26,565)}$$

$$\Delta l = \bar{u}_D \cos \alpha = \bar{u}_D \cos(26,565)$$

$$\begin{aligned} \bar{u}_D = \cancel{\bar{u}_A} + \bar{\theta}_A \cdot 1 + \frac{1}{EI} \left[+ \frac{1}{2} \cdot 200 \cdot 2 \cdot \left(\frac{2}{3} \cdot 2 - 1\right) + 100 \cdot 1 \cdot 1 \right. \\ \left. + \frac{1}{2} \cdot 100 \cdot 1 \cdot 1 + \frac{1}{2} \cdot 100 \cdot \sqrt{2} \cdot \left(\frac{2}{3}\right) - \frac{1}{2} \cdot 1,788 \cdot F_c \cdot 2 \cdot \left(2 \cdot \frac{2}{3} - 1\right) \right. \\ \left. - 1,3407 \cdot F_c \cdot 1 \cdot 1 - \frac{1}{2} \cdot 0,4473 F_c \cdot 1 \cdot 1 - \frac{1}{2} \cdot 1,3407 F_c \cdot \sqrt{2} \cdot \frac{2}{3} \right] \end{aligned}$$

$$\vec{UD} = \vec{\theta A} + 8,79 \cdot 10^{-3} - 9,307 \cdot 10^{-5} F_c \quad (3)$$

$$\begin{cases} 2 \frac{F_c}{EA} \cdot \frac{1}{\cos^2(26,565)} = \vec{\theta A} + 8,79 \cdot 10^{-3} - 9,307 \cdot 10^{-5} F_c & (2,3) \\ 2 \vec{\theta A} + 2,197 \cdot 10^{-4} F_c - 0,0227 = 0 & (1) \end{cases}$$

$$\begin{cases} 7,1807 \cdot 10^{-4} F_c - \vec{\theta A} = 8,79 \cdot 10^{-3} \\ 2,197 \cdot 10^{-4} F_c + 2 \vec{\theta A} = 0,0227 \end{cases} \Rightarrow \begin{aligned} F_c &= 24,326 \text{ kN} \\ \theta A &= 8,677 \cdot 10^{-3} \text{ rad} \end{aligned}$$

b) Las leyes de esfuerzos

Reacciones:

$$\begin{aligned} V_A &= -100 \text{ kN} \\ H_A &= -100 \text{ kN} \\ V_D &= 100 \text{ kN} \end{aligned}$$

